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Mapping Global Talent Pipelines in Biosecurity

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EXECUTIVE SUMMARY

The Global South needs context-specific, up-to-date talent building programmes to address recent biorisks. We mapped 201 programmes across 110+ countries and identified three structural gaps: pipelines target existing practitioners, syllabi omit AI × Bio and DNA-synthesis screening, and in-person formats exclude participants from the Global South.

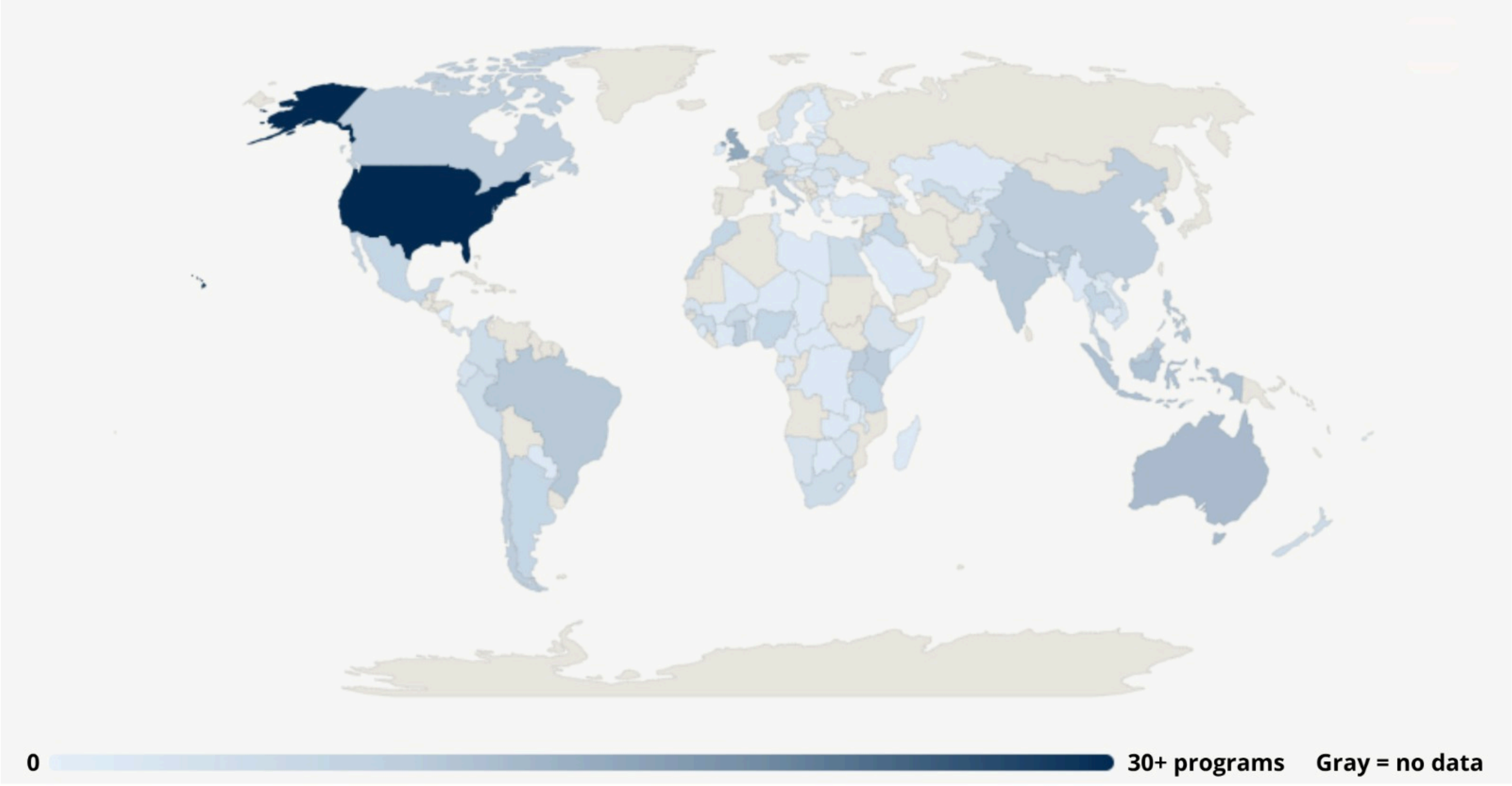
BACKGROUND & CONTEXT

AI tools, frontier LLMs, and accessible DNA synthesis are eroding barriers that once limited bioweapon development, yet biosafety and biosecurity remain among the weakest core capacities under the International Health Regulations. The field is largely invisible to students entering the life sciences, lacks a clear career roadmap, and has historically been built on Global North standards and donor cycles that leave Global South pipelines unsustainable.

METHODS

1. Systematic search across 9 global regions × 3 programme categories → 27 targeted Gemini searches.
2. Four-stage pipeline: ingestion → classification → extraction → deduplication, yielding a curated dataset of 201 programmes.
3. Data analysis of geographic distribution, programme structure, audience reach, and curricular coverage.

GLOBAL DISTRIBUTION



Programme density by country. Training programmes concentrate in a handful of Global North countries; most of the Global South (including regions most exposed to emerging biorisk) host few or no programmes.

RECOMMENDATIONS

01 Integrate AI × Bio and DNA-synthesis screening into existing syllabi.

Of thirteen AI × Bio programmes, ten are fellowships or competitions. Revise existing biosafety courses to include AI × Bio and synthesis-screening modules, and use low-friction online hackathons and short courses as entry points for Global South audiences.

02 Expand hybrid delivery; underwrite in-person access.

In-person delivery accounts for 58% of all programmes, imposing travel, visa, and employment-leave costs that fall disproportionately on Global South participants. Convert existing in-person programmes to hybrid, and pair remaining in-person formats with visa and travel support.

03 Recruit early-career talent through dedicated entry pathways.

Set explicit undergraduate and early-career targets in funding calls. Reserve cohort slots in fellowships for participants without prior biosecurity experience.

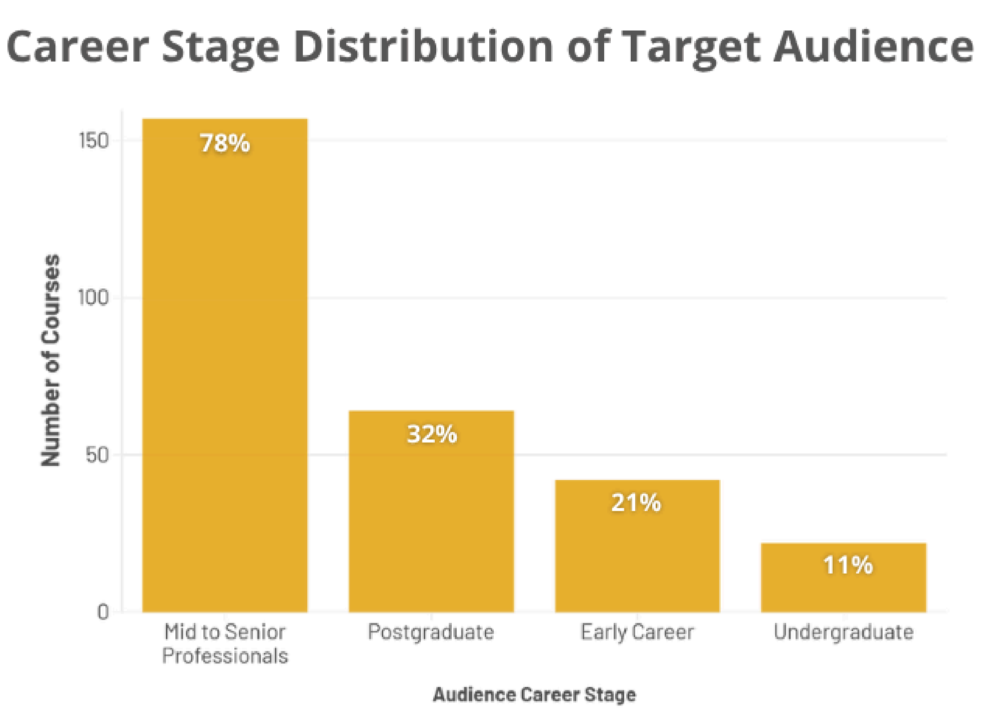
04 Invest in formal Global South degree programmes as a long-term career anchor.

Formal degrees embed biosecurity within national higher education systems and provide an institutional base that does not rely on continuous external funding. Replicate models like the IFBA–MMUST undergraduate biosafety degree in Kenya to embed biosecurity in national education systems and reduce funding-cycle attrition.

05 Reorient donor funding toward multi-year, locally co-designed programmes.

Multilateral and government-led programmes are 3× more common in the Global South (21% vs 7%), so external donors shape much of Global South training. Commit to multi-year horizons, co-design curricula with local institutions, and build transition plans that hand operational ownership to local actors.

RESULTS · AUDIENCE, DELIVERY & CURRICULAR GAP



Most programmes target mid- to senior-level professionals; only 11% reach undergraduates.

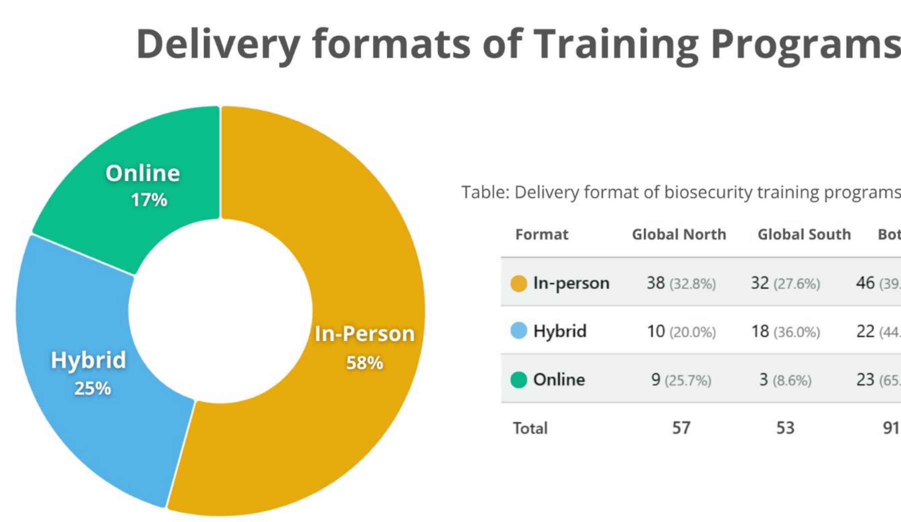
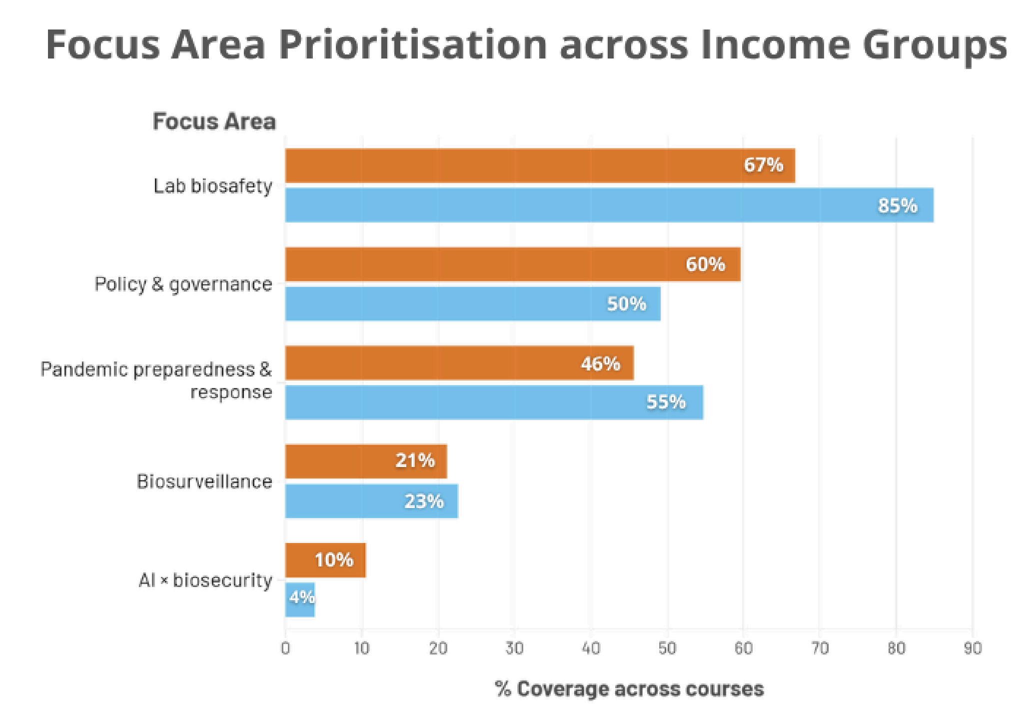


Table: Delivery format of biosecurity training programs by income group

Format	Global North	Global South	Both	Total
In-person	38 (32.8%)	32 (27.6%)	46 (39.7%)	116
Hybrid	10 (20.0%)	18 (36.0%)	22 (44.0%)	50
Online	9 (25.7%)	3 (8.6%)	23 (65.7%)	35
Total	57	53	91	201

Delivery formats aren't accessible: 58% of programmes are in-person, layering travel and visa costs onto Global South participants.



Emerging biorisks remain a curricular blind spot: just 10% of Global North and 4% of Global South programmes engage AI × Bio, and DNA-synthesis screening is virtually absent from either.